Chapter 7
Perfect Competition and the Supply Curve

What you will learn in this chapter:

- The meaning of **perfect competition** and the characteristics of a **perfectly competitive industry**
- The difference between **accounting profit** and **economic profit**, and why economic profit is the correc

basis for decisions.

- How a **price-taking producer** determines its profit-maximizing quantity of output
- **Deriving supply curve from the profit maximization**
- **Deriving industry supply curve from individual supply curves**
- Why industries behave differently in the short run and the long run
- What determines the **industry supply curve** in both the short run and the long run

Perfect Competition

Perfectly Competitive Market

All market participants, both consumers and producers, are price-takers.

(The price is determined by the **industry demand** and the **industry supply**.)

A **price-taking producer** is a producer whose actions have no effect on the market price of the good it sells.

A **price-taking consumer** is a consumer whose actions have no effect on the market price of the good he or she buys.

A **perfectly competitive industry** is an industry in which producers are price takers.

Perfect Competition

Two Necessary Conditions for Perfect Competition

1. None of producers have a large market share. A producer's **market share** is the fraction of the total industry output represented by that producer's output.

2. A good is a **standardized product**, also known as a **commodity**, when consumers regard the products of different producers as the same good.

Free Entry and Exit

There is **free entry and exit** into and from an industry when new producers can easily enter into or leave that industry.

Profit Maximization

Profit = Total Revenue - Total Cost.

$\pi(Q) = TR(Q) - TC(Q)$,

where $\pi(Q)$ is the profit function,

$TR(Q)$, $TC(Q)$ is the Total Revenue, Total Cost function.

Each function depends on how much the firm produces (sells).

To the $\pi(Q)$ be maximum,

Marginal Revenue = Marginal Cost

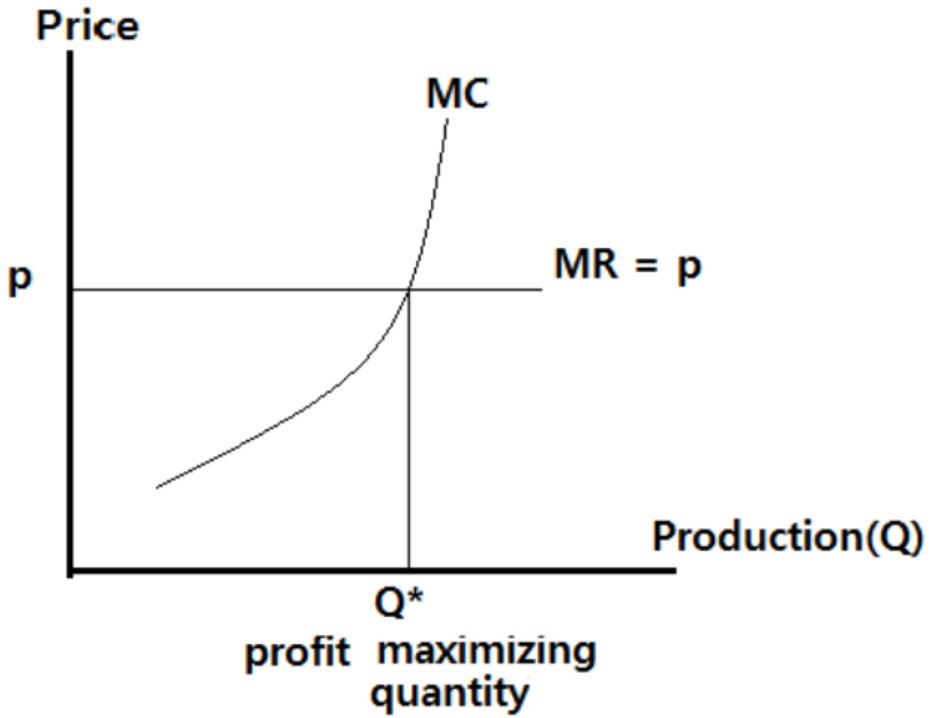
In the perfect competition market, MR = p

So $\frac{d\pi(Q)}{dQ} = 0;\ \frac{dTR(Q)}{dQ} - \frac{dTC(Q)}{dQ} = 0$ nder the perfect competition.

When MR = MC profit(π) becomes the maximum

(Q)	(MR)	(TR)	(FC)	(VC)	(TC)	(MC)	(π)
0	18	0	14	0	14	16	-14
1	18	18	14	16	30	4	-12
2	18	36	14	20	34	5	2
3	18	54	14	25	39	17	15
4	18	72	14	42	56	18	16
5	18	90	14	60	74	40	16
6	18	108	14	100	114	30	-6
7	18	126	14	130	144		

Production and Profits



Production and Profits

When Is Production Profitable?

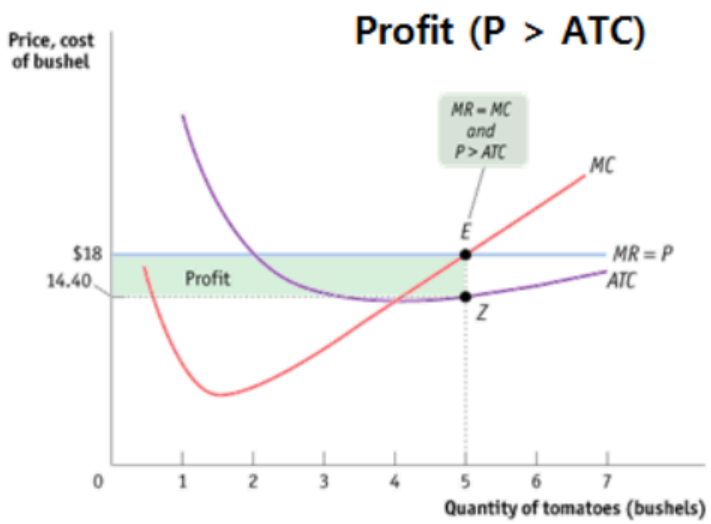
- If $P > ATC$

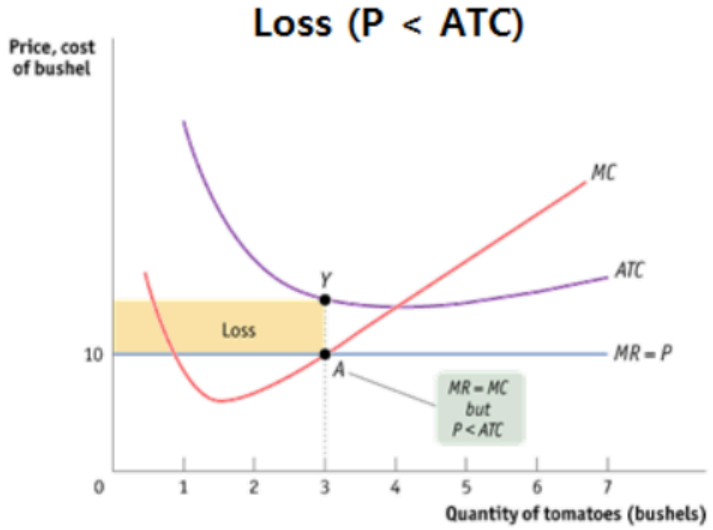
(when the price is higher than average total cost, the firm is profitable.)

- If $P = ATC$, the firm breaks even.
- If $P < ATC$, the firm incurs a loss.

Production and Profits

When Is Production Profitable?





Production and Profits

When Is Production Profitable?

The **break-even price** of a price-taking firm is the market price at which it earns zero profits.

The rule for determining whether a producer of a good is profitable depends on a comparison of the market price of the good to the producer's break-even price—its minimum average total cost:

- Whenever the market price exceeds minimum average total cost, the producer is profitable.
- Whenever the market price equals minimum average total cost, the producer breaks even.
- Whenever the market price is less than minimum average total cost, the producer is unprofitable.

Should I operate my business even if I get loss?

	Fixed Cost	Variable Cost	Total Cost	Revenue	Profit	Operate?
a.	1000	2000	3000	5000	2000	Yes
b.	1000	3000	4000	4000	0	Yes
c.	1000	4000	5000	4500	-500	Yes
d.	1000	5000	6000	5000	-1000	Maybe
e.	1000	6000	7000	5500	-1500	No

Case c:

Why you keep operating the business when you lose your money?

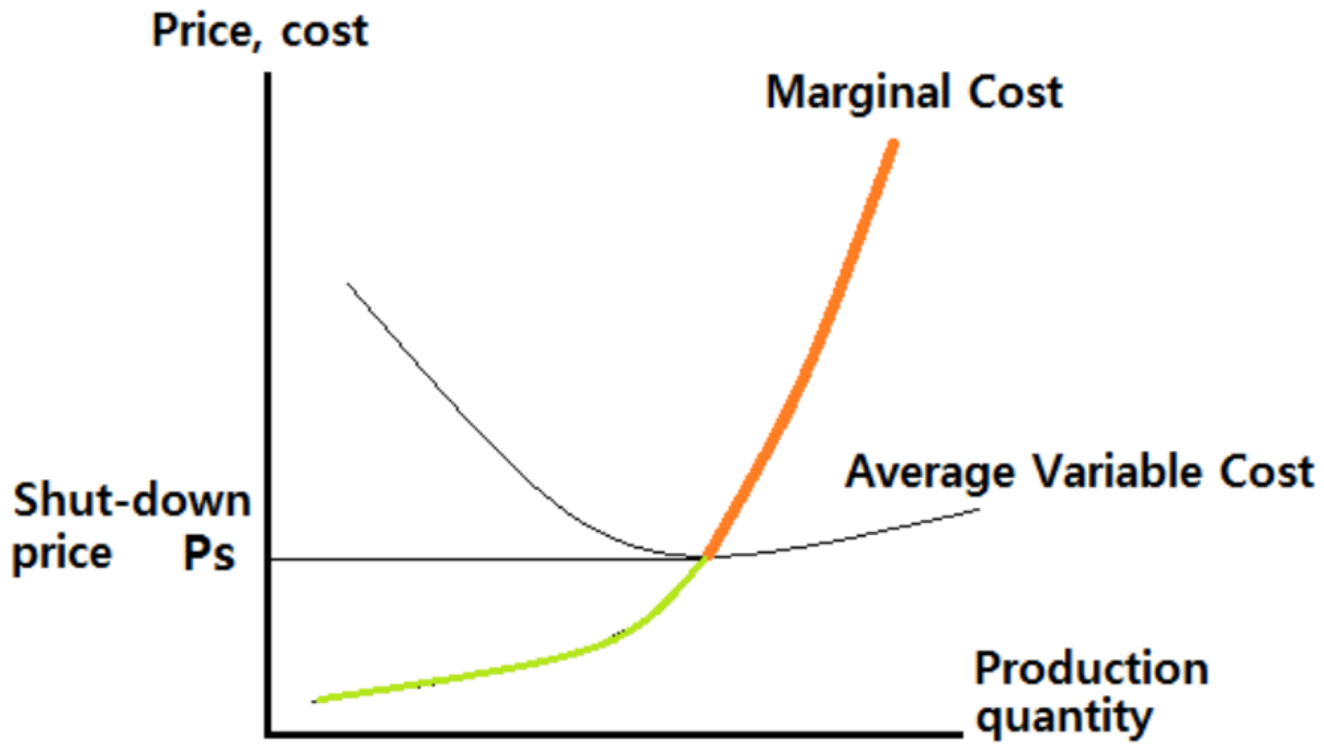
If you shut down, you will lose more money because of the fixed cost.

For example, in the case c, you have to pay 1000 even if you shut down the business. However, if you operate, you will lose only 500.

Business Operating Condition: Until $-(TR - VC - FC) < FC$

Since $TR = P*Q$, $VC = Q* AVC$, it becomes $P > AVC$

The Short-Run Individual Supply Curve



The short-run supply curve is the part of the marginal cost which satisfies
P > Ps , Ps: minimum average cost

Summary

1. A firm maximizes profit at the production point where the MR = MC condition is satisfied.

1. In the perfect competition industry, individual firm's marginal revenue is P.

1. So the maximum profit condition of perfect competition industry is P = MC

2. Therefore individual firm's supply curve is its marginal cost curve

3. However, a firm will shut down when the market price is less than the average variable cost.

4. So the supply curve is part of marginal curve where P > AVC condition meets.

Production and Profits

Changing Fixed Cost

Although fixed cost cannot be altered in the short run, in the long run the level of fixed cost is a matter of choice.

In most perfectly competitive industries the set of producers, although fixed in the short run, changes in the long run as firms enter or leave the industry.

Production and Profits

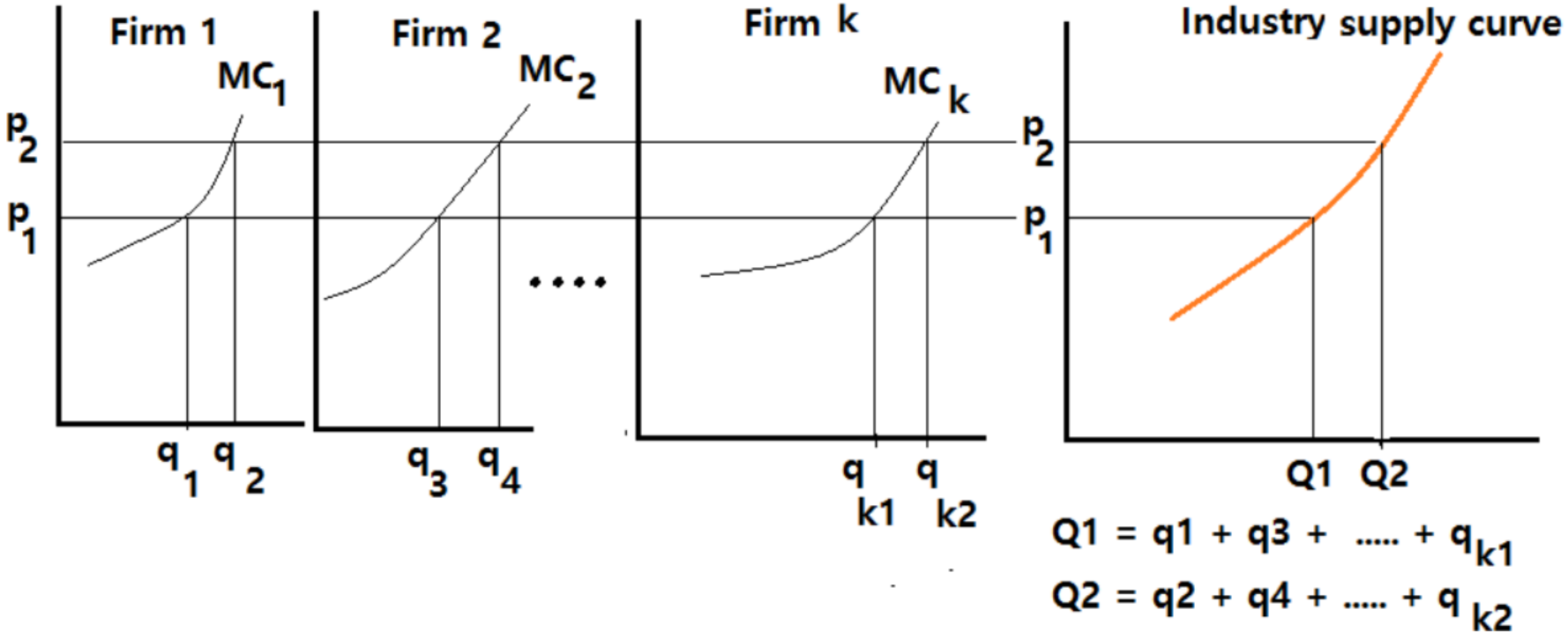
Summing Up: The Perfectly Competitive Firm's Profitability and Production Conditions

TABLE 7-4 Summary of the Perfectly Competitive Firm's Profitability and Production Conditions

Profitability condition (minimum *ATC* = break-even price)	**Result**
$P >$ minimum *ATC*	Firm profitable. Entry into industry in the long run.
$P =$ minimum *ATC*	Firm breaks even. No entry into or exit from industry in the long run.
$P <$ minimum *ATC*	Firm unprofitable. Exit from industry in the long run.
Production condition (minimum *AVC* = shut-down price)	**Result**
$P >$ minimum *AVC*	Firm produces in the short run. If $P <$ minimum *ATC*, firm covers variable cost and some but not all of fixed cost. If $P >$ minimum *ATC*, firm covers all variable cost and fixed cost.
$P =$ minimum *AVC*	Firm indifferent between producing in the short run or not. Just covers variable cost.
$P <$ minimum *AVC*	Firm shuts down in the short run. Does not cover variable cost.

From the Individual supply curves to the Industry Supply Curve

The **industry supply curve** shows the relationship between the price of a good and the total output of the industry as a whole.



The Industry Supply Curve

The Short-Run Industry Supply Curve

The **short-run industry supply curve** shows how the quantity supplied by an industry depends on the market price, given a fixed number of producers.

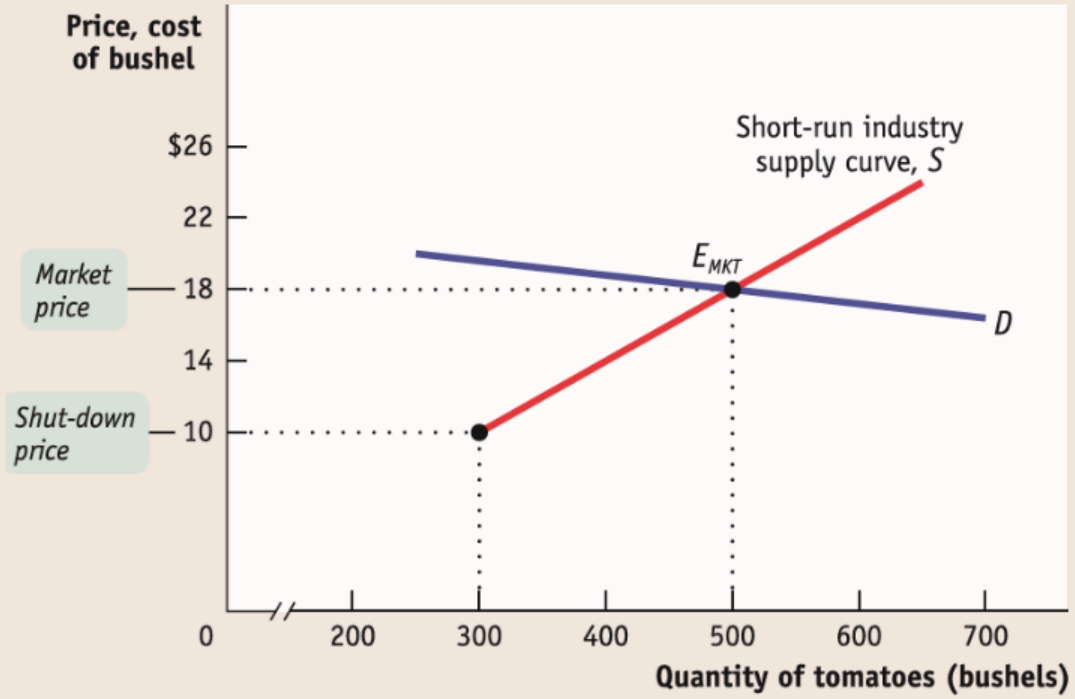
There is a **short-run market equilibrium** when the quantity supplied equals the quantity demanded, taking the number of producers as given.

The Industry Supply Curve

The Short-Run Industry Supply Curve

FIGURE 7-5 The Short-Run Market Equilibrium

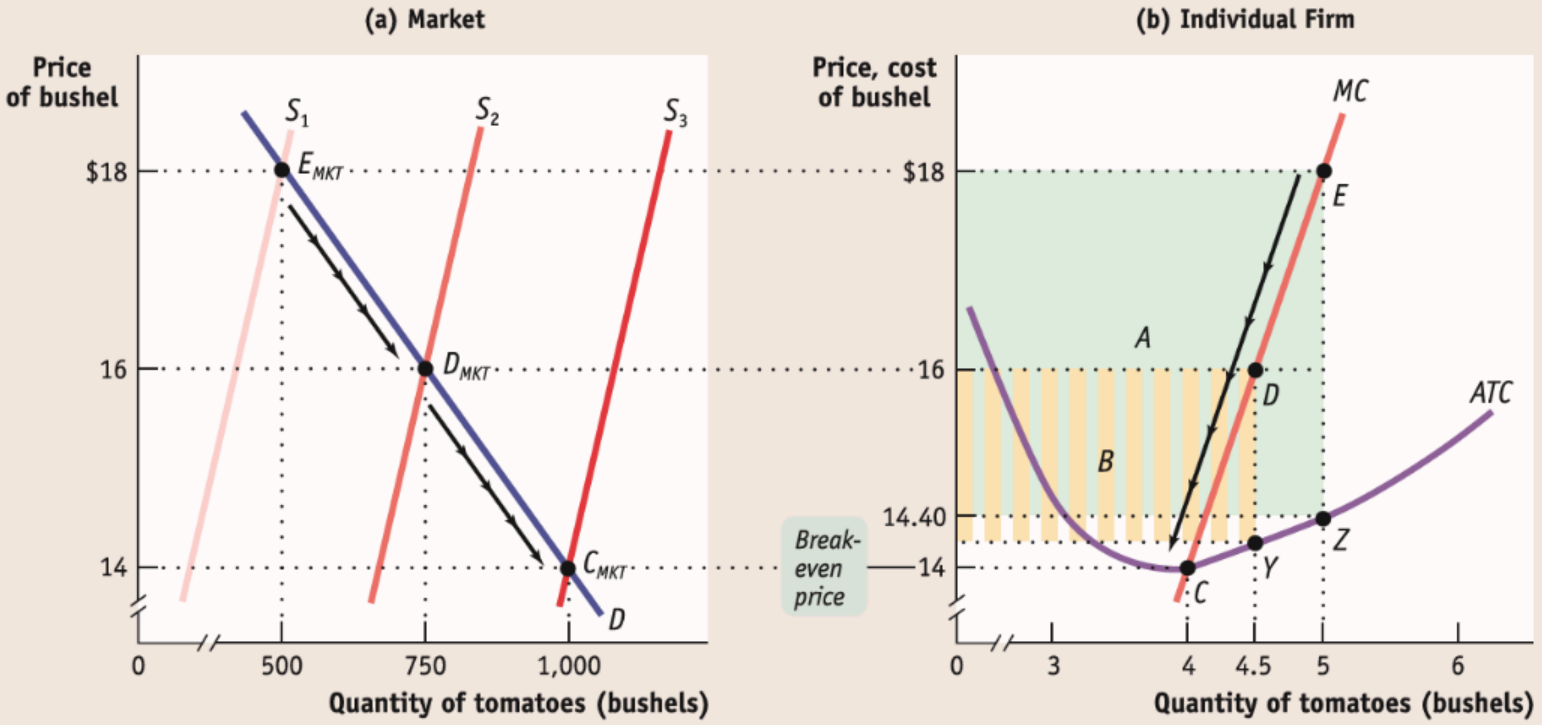
The short-run industry supply curve, *S*, is the industry supply curve taking the number of producers—here, 100—as given. It is generated by adding together the individual supply curves of the 100 producers. Below the shut-down price of $10, no producer wants to produce in the short run. Above $10, the short-run industry supply curve slopes upward, as each producer increases output as price increases. It intersects the demand curve, *D*, at point E_{MKT}, the point of short-run market equilibrium, corresponding to a market price of $18 and a quantity of 500 bushels.



The Industry Supply Curve

The Long-Run Industry Supply Curve

FIGURE 7-6 The Long-Run Market Equilibrium



Point E_{MKT} of panel (a) shows the initial short-run market equilibrium. Each of the 100 existing producers makes an economic profit, illustrated in panel (b) by the green rectangle labeled *A*, the profit of an existing firm. Profits induce entry by additional producers, shifting the short-run industry supply curve outward from S_1 to S_2 in panel (a), resulting in a new short-run equilibrium at point D_{MKT}, at a lower market price of $16 and higher industry output. Existing firms reduce output and profit falls to the area given by the striped rectangle labeled *B* in panel (b). Entry continues to shift out the short-run industry supply curve, as price falls and industry output increases yet again. Entry ceases at point C_{MKT} on supply curve S_3 in panel (a). Here market price is equal to the break-even price; existing producers make zero economic profits, and there is no incentive for entry or exit. So C_{MKT} is also a long-run market equilibrium.

The Industry Supply Curve

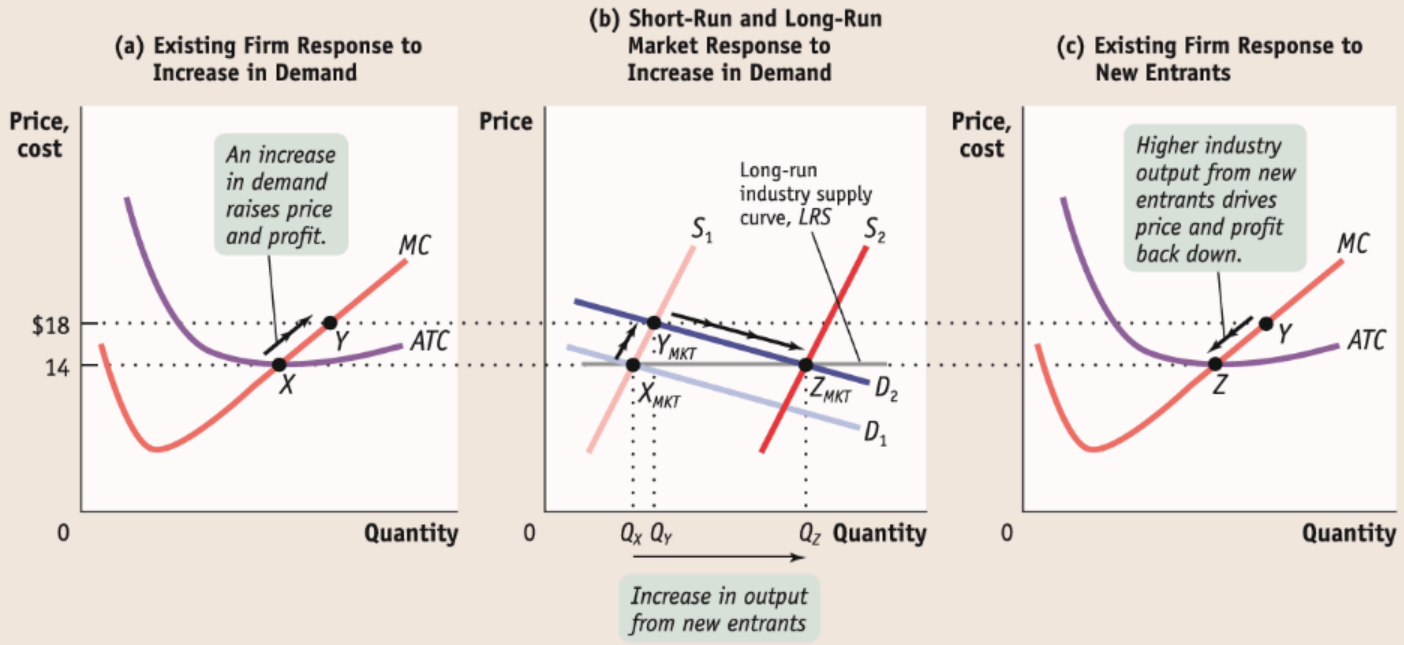
The Long-Run Industry Supply Curve

A market is in **long-run market equilibrium** when the quantity supplied equals the quantity demanded, given that sufficient time has elapsed for entry into and exit from the industry to occur.

The Industry Supply Curve

The Long-Run Industry Supply Curve

FIGURE 7-7 The Effect of an Increase in Demand in the Short Run and the Long Run



Panel (b) shows how an industry adjusts in the short and long run to an increase in demand; panels (a) and (c) show the corresponding adjustments by an existing firm. Initially the market is at point X_{MKT} in panel (b), a short-run and long-run equilibrium at a price of $14 and industry output of Q_X. An existing firm makes zero economic profit, operating at point X in panel (a) at minimum average total cost. Demand increases as D_1 shifts rightward to D_2 in panel (b), raising the market price to $18. Existing firms increase their output, and industry output moves along the short-run industry supply curve S_1 to a short-run equilibrium at Y_{MKT}. Correspondingly, the existing firm in panel (a) moves from point X to point Y. But at a price of $18 existing firms are profitable. As shown in panel (b), in the long run new entrants arrive and the short-run industry supply curve shifts rightward, from S_1 to S_2. There is a new equilibrium at point Z_{MKT}, at a lower price of $14 and higher industry output of Q_Z. An existing firm responds by moving from Y to Z in panel (c), returning to its initial output level and zero economic profit. Production by new entrants accounts for the total increase in industry output, Q_Z Q_X. Like X_{MKT}, Z_{MKT} is also a short-run and long-run equilibrium: with existing firms earning zero economic profit, there is no incentive for any firms to enter or exit the industry. The horizontal line passing through X_{MKT} and Z_{MKT}, *LRS*, is the long-run industry supply curve: at the break-even price of $14, producers will produce any amount that consumers demand in the long run.

The Industry Supply Curve

The Long-Run Industry Supply Curve

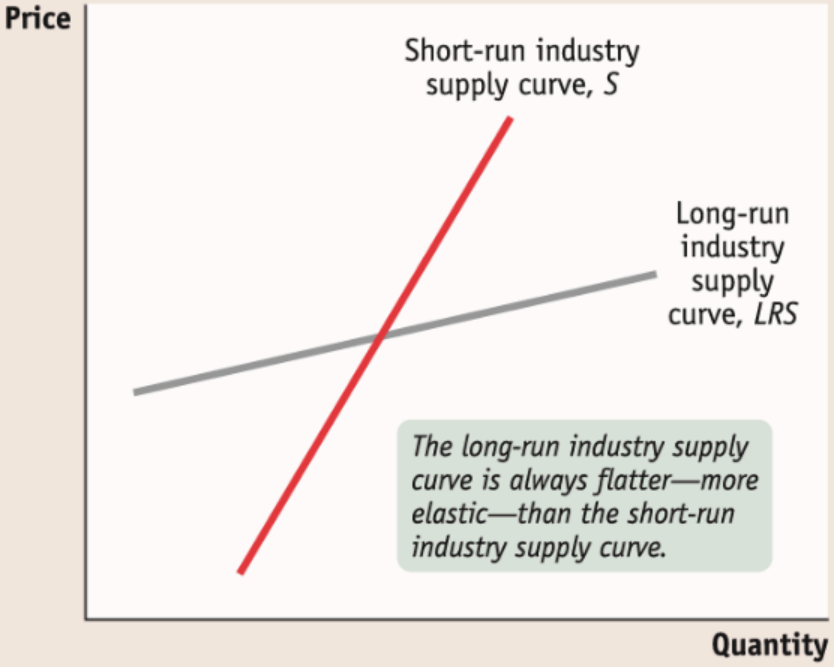
The **long-run industry supply curve** shows how the quantity supplied responds to the price once producers have had time to enter or exit the industry.

The Industry Supply Curve

The Long-Run Industry Supply Curve

FIGURE 7-8 Comparing the Short-Run and Long-Run Industry Supply Curves

The long-run industry supply curve may slope upward, but it is always flatter—more elastic—than the short-run industry supply curve. This is because of entry and exit: a higher price attracts new entrants in the long run, resulting in a rise in industry output and a fall in price; a lower price induces existing producers to exit in the long run, generating a fall in industry output and an eventual rise in price.



The Industry Supply Curve

The Cost of Production and Efficiency in Long-run Equilibrium

Our analysis leads us to three conclusions about the cost of production and efficiency in the long-run equilibrium of a perfectly competitive industry.

First, in a perfectly competitive industry in equilibrium, the value of marginal cost is the same for all firms.

Second, in a perfectly competitive industry with free entry and exit, each firm will have zero economic profit in long-run equilibrium.

The third and final conclusion is that the long-run market equilibrium of a perfectly competitive industry is efficient: no mutually beneficial transactions go unexploited.

KEY TERMS

Price-taking producer
Price-taking consumer
Perfectly competitive market
Perfectly competitive industry
Market share
Standardized product
Commodity
Free entry and exit
Principle of marginal analysis
Marginal revenue
Optimal output rule
Price-taking firm's optimal output rule
Marginal revenue curve

Accounting profit
Economic profit
Capital
Implicit cost of capital
Break-even price
Shut-down price
Sunk costs
Short-run individual supply curve
Industry supply curve
Short-run industry supply curve
Short-run market equilibrium
Long-run market equilibrium
Long-run industry supply curve